

rootlog

November 5, 2025

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[3]: import networkx as nx
import matplotlib.pyplot as plt
import numpy as np
import time
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[10]: class graphWithComplement:

    def __init__(self, n, p=None):
        self.n = n
        self.half_of_all_edges = n*(n-1)//4
        if p is None:
            self.graph = nx.empty_graph(n)
            self.complement = nx.complete_graph(n)
        else:
            self.graph = nx.fast_gnp_random_graph(n, p)
            self.complement = nx.complement(self.graph)

    def add_edge(self, u, v):
        self.graph.add_edge(u, v)
        self.complement.remove_edge(u, v)

    def remove_edge(self, u, v):
        self.graph.remove_edge(u, v)
        self.complement.add_edge(u, v)

    def has_edge(self, u, v):
        return self.graph.has_edge(u, v)

    def number_of_common_neighbors(self, u, v, m=None):
        if m is None or m < self.half_of_all_edges:
            return len(nx.common_neighbors(self.graph, u, v))
        else:
            return self.n - 2 - len((self.complement.adj[u].keys() | self.
↪complement.adj[v].keys()) - {u,v})
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[13]: n = 500
alpha = 1 #lagrangian multiplier for sum of P = 1 constraint
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theta1 = 2.2
G = graphWithComplement(n, p=0.5)
m= G.graph.number_of_edges()
t = sum(nx.triangles(G.graph).values()) // 3
its = 500000
burn_in = 50000
up_edges = []
up_tris = []
down_edges = []
down_tris = []

#upsweep
theta2_vals = np.linspace(0, 5, 100)
for theta2 in theta2_vals:
    H_old_sq = (alpha + theta1 * m - theta2 * t) ** 2
    start_time = time.time()
    total_edges = 0
    total_tris = 0

    for _ in range(its):
        i, j = np.random.randint(0, n, 2)
        while(i==j):
            i, j = np.random.randint(0, n, 2)
        if G.has_edge(i, j):
            delta_t = -G.number_of_common_neighbors(i, j, m)
            delta_m = -1
            G.remove_edge(i, j)
        else:
            delta_t = G.number_of_common_neighbors(i, j, m)
            delta_m = 1
            G.add_edge(i, j)

        t_new = t + delta_t
        m_new = m + delta_m
        H_new_sq = (alpha + theta1 * m_new - theta2 * t_new) ** 2

    dH = H_old_sq - H_new_sq
    if np.log(np.random.uniform()) <= dH:
        m += delta_m
        t += delta_t
        H_old_sq = H_new_sq
    else:
        # revert
        if delta_m == -1:
            G.add_edge(i, j)
        else:
            G.remove_edge(i, j)

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        if _ >= burn_in:
            total_edges += m
            total_tris += t

    elapsed = time.time() - start_time
    print(f"theta2 = {theta2:.4f}, time = {elapsed:.2f} s")

    up_edges.append(total_edges / (its-burn_in))
    up_tris.append(total_tris / (its-burn_in))

#downsweep
for theta2 in reversed(theta2_vals):
    H_old_sq = (alpha + theta1 * m - theta2 * t) ** 2
    start_time = time.time()
    total_edges = 0
    total_tris = 0

    for _ in range(its):
        i, j = np.random.randint(0, n, 2)
        while(i==j):
            i, j = np.random.randint(0, n, 2)
        if G.has_edge(i, j):
            delta_t = -G.number_of_common_neighbors(i, j, m)
            delta_m = -1
            G.remove_edge(i, j)
        else:
            delta_t = G.number_of_common_neighbors(i, j, m)
            delta_m = 1
            G.add_edge(i, j)

        t_new = t + delta_t
        m_new = m + delta_m
        H_new_sq = (alpha + theta1 * m_new - theta2 * t_new) ** 2

    dH = H_old_sq - H_new_sq
    if np.log(np.random.uniform()) <= dH:
        m += delta_m
        t += delta_t
        H_old_sq = H_new_sq
    else:
        # revert
        if delta_m == -1:
            G.add_edge(i, j)
        else:
            G.remove_edge(i, j)

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        if _ >= burn_in:
            total_edges += m
            total_tris += t

    elapsed = time.time() - start_time
    print(f"theta2 = {theta2:.4f}, time = {elapsed:.2f} s")

    down_edges.append(total_edges / (its-burn_in))
    down_tris.append(total_tris / (its-burn_in))

plt.figure()
plt.scatter(theta2_vals, up_edges, label='upsweep', marker = 'o')
plt.scatter(theta2_vals[::-1], down_edges, label = 'downsweep', marker = '^')

plt.figure()
plt.scatter(theta2_vals, up_tris, label='upsweep', marker = 'o')
plt.scatter(theta2_vals[::-1], down_tris, label = 'downsweep', marker = '^')

plt.show()

```

```

theta2 = 0.0000, time = 17.25 s
theta2 = 0.0505, time = 11.03 s
theta2 = 0.1010, time = 11.82 s
theta2 = 0.1515, time = 11.81 s
theta2 = 0.2020, time = 10.78 s
theta2 = 0.2525, time = 11.79 s
theta2 = 0.3030, time = 11.92 s
theta2 = 0.3535, time = 10.88 s
theta2 = 0.4040, time = 11.81 s
theta2 = 0.4545, time = 11.98 s
theta2 = 0.5051, time = 11.04 s
theta2 = 0.5556, time = 11.91 s
theta2 = 0.6061, time = 11.47 s
theta2 = 0.6566, time = 12.12 s
theta2 = 0.7071, time = 12.04 s
theta2 = 0.7576, time = 12.32 s
theta2 = 0.8081, time = 11.51 s
theta2 = 0.8586, time = 11.57 s
theta2 = 0.9091, time = 10.59 s
theta2 = 0.9596, time = 11.54 s
theta2 = 1.0101, time = 10.60 s
theta2 = 1.0606, time = 11.50 s
theta2 = 1.1111, time = 609.35 s
theta2 = 1.1616, time = 9.89 s
theta2 = 1.2121, time = 8.27 s
theta2 = 1.2626, time = 7.65 s
theta2 = 1.3131, time = 8.46 s

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theta2 = 1.3636, time = 9.50 s
theta2 = 1.4141, time = 9.34 s
theta2 = 1.4646, time = 8.50 s
theta2 = 1.5152, time = 8.96 s
theta2 = 1.5657, time = 8.55 s
theta2 = 1.6162, time = 7.76 s
theta2 = 1.6667, time = 8.54 s
theta2 = 1.7172, time = 9.02 s
theta2 = 1.7677, time = 9.53 s
theta2 = 1.8182, time = 8.12 s
theta2 = 1.8687, time = 9.25 s
theta2 = 1.9192, time = 10.01 s
theta2 = 1.9697, time = 8.84 s
theta2 = 2.0202, time = 9.03 s
theta2 = 2.0707, time = 9.07 s
theta2 = 2.1212, time = 9.05 s
theta2 = 2.1717, time = 8.32 s
theta2 = 2.2222, time = 9.09 s
theta2 = 2.2727, time = 9.08 s
theta2 = 2.3232, time = 8.19 s
theta2 = 2.3737, time = 9.15 s
theta2 = 2.4242, time = 9.07 s
theta2 = 2.4747, time = 9.03 s
theta2 = 2.5253, time = 8.25 s
theta2 = 2.5758, time = 9.15 s
theta2 = 2.6263, time = 9.34 s
theta2 = 2.6768, time = 8.82 s
theta2 = 2.7273, time = 9.50 s
theta2 = 2.7778, time = 9.47 s
theta2 = 2.8283, time = 8.94 s
theta2 = 2.8788, time = 8.48 s
theta2 = 2.9293, time = 9.15 s
theta2 = 2.9798, time = 9.10 s
theta2 = 3.0303, time = 8.32 s
theta2 = 3.0808, time = 9.15 s
theta2 = 3.1313, time = 8.96 s
theta2 = 3.1818, time = 9.00 s
theta2 = 3.2323, time = 8.28 s
theta2 = 3.2828, time = 9.01 s
theta2 = 3.3333, time = 9.08 s
theta2 = 3.3838, time = 8.92 s
theta2 = 3.4343, time = 8.23 s
theta2 = 3.4848, time = 8.96 s
theta2 = 3.5354, time = 8.91 s
theta2 = 3.5859, time = 8.14 s
theta2 = 3.6364, time = 8.81 s
theta2 = 3.6869, time = 8.94 s
theta2 = 3.7374, time = 9.03 s

theta2 = 3.7879, time = 8.05 s
theta2 = 3.8384, time = 9.06 s
theta2 = 3.8889, time = 9.00 s
theta2 = 3.9394, time = 7.98 s
theta2 = 3.9899, time = 9.02 s
theta2 = 4.0404, time = 8.99 s
theta2 = 4.0909, time = 8.87 s
theta2 = 4.1414, time = 7.98 s
theta2 = 4.1919, time = 8.94 s
theta2 = 4.2424, time = 9.01 s
theta2 = 4.2929, time = 9.10 s
theta2 = 4.3434, time = 7.91 s
theta2 = 4.3939, time = 8.85 s
theta2 = 4.4444, time = 8.94 s
theta2 = 4.4949, time = 8.21 s
theta2 = 4.5455, time = 8.94 s
theta2 = 4.5960, time = 9.06 s
theta2 = 4.6465, time = 8.93 s
theta2 = 4.6970, time = 7.99 s
theta2 = 4.7475, time = 8.98 s
theta2 = 4.7980, time = 8.95 s
theta2 = 4.8485, time = 8.14 s
theta2 = 4.8990, time = 8.99 s
theta2 = 4.9495, time = 8.99 s
theta2 = 5.0000, time = 9.01 s
theta2 = 5.0000, time = 8.00 s
theta2 = 4.9495, time = 8.98 s
theta2 = 4.8990, time = 8.97 s
theta2 = 4.8485, time = 9.00 s
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theta2 = 2.4747, time = 8.49 s
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theta2 = 0.7576, time = 8.55 s
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theta2 = 0.3030, time = 7.31 s
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theta2 = 0.0000, time = 8.40 s



